

Module 1: Network + Network Fundamentals and building networks

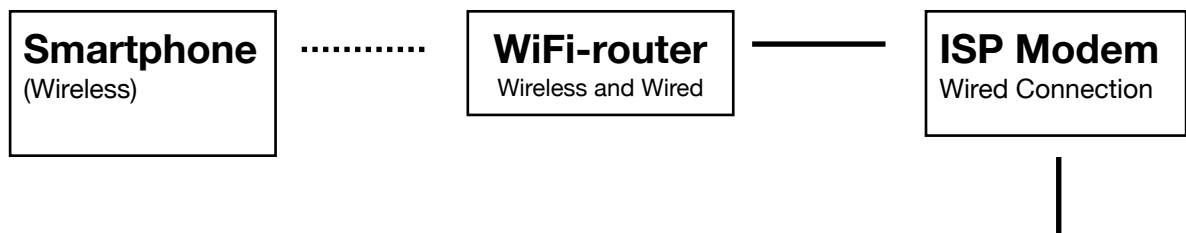
Wireless Networking

Task 1:

Three popular Wi-Fi standards

- **802.11n (Wi-Fi 4)** — Max speed around 600 Mbps, operates on **both 2.4 GHz and 5 GHz** bands, introduced MIMO (multiple antennas) to boost range and speed over older standards.
- **802.11ac (Wi-Fi 5)** — Max speed can exceed 1 Gbps, operates **only on the 5 GHz band**, using wider channels and MU-MIMO to support more devices simultaneously at higher speed than 802.11n.
- **802.11ax (Wi-Fi 6)** — Max speed can reach several Gbps, works on **both 2.4 GHz and 5 GHz** (Wi-Fi 6E extends into 6 GHz too), and is designed specifically to handle many connected devices at once more efficiently — better for crowded homes/offices than 802.11ac.

Task 2:



Task 4:

SSID hiding

SSID hiding means configuring your router to not broadcast its network name publicly, so it won't show up in the list of available Wi-Fi networks that nearby devices scan for — anyone wanting to connect has to already know the exact SSID and type it in manually.

- **Advantage:** It adds a small layer of obscurity — casual, non-technical passersby won't even see your network exists, which can reduce opportunistic connection attempts.

- **Disadvantage:** It doesn't provide real security — anyone using basic Wi-Fi scanning tools can still detect the hidden network's presence, and it actually makes connecting your own legitimate devices more annoying, since you have to manually enter the SSID each time instead of just picking it from a list.

